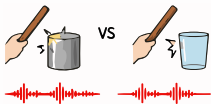


Physically-Grounded Acoustic Scenarios

— Generation Stage —

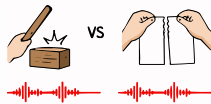
Material Property Inference

Metal vs Glass



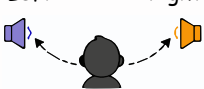
Excitation Mechanism Recognition

Hit vs Tear



Sound Source Localization

Left vs Right



Source Motion Inference

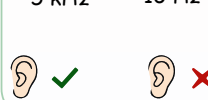
Approach vs Move away



— Perception Stage —

Audibility Detection

5 kHz vs 10 Hz



Loudness Perception Inference

High frequency vs Low frequency



Ambiguous Physical Interpretation

What material is it?

How was it produced?

Where is it?

Is it moving?

How far away is it?

What's the environment?



I struggle to infer physical causes from sound!

Audio LLM

- ✗ Weak in reasoning and calculation
- ✗ Unable to understand the world



Our Goal: Audio LLMs can perceive, reason and understand the physical world through sound.



Better World Understanding



More Accurate Reasoning



Closer to Human Perception